Pergamon

PII: S0094-5765(96)00100-2

Acta Astronautica Vol. 39, No. 5, pp. 381–394, 1996
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0094-5765/96 $15.00 + 0.00

LUNAR BASE THERMAL CONTROL SYSTEMS USING HEAT PUMPS†

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(Received 7 June 1995)

Abstract—There is considerable interest at the present time to design a thermal control system (TCS) for a lunar base. Conventional techniques cannot be used for the purpose due to the lack of a readily available temperature sink during most of the lunar day. The lunar surface near the equatorial regions reaches a maximum of about 390 K during the 336-h lunar day. The projected range of temperatures for operation of sensors and conditioned habitat spaces is 270–293 K. A heat pump augmented TCS can be used to increase the operating temperature of the radiator, thereby enabling heat rejection. Rankine, absorption, and reverse Brayton cycle heat pumps with various working fluids are examined to identify the optimal cycle and working fluid combination. A base-line cooling load of 100 kW to be rejected at 270 K is used in the analysis. A Rankine cycle heat pump operating with R11 as the working fluid and R717 as a rejection loop coolant provides an optimal total TCS mass of 5940 kg at a radiator temperature of 362 K.
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1. INTRODUCTION

The design of a thermal control system that is capable of cooling a permanently inhabited lunar base is discussed here. Moderate-temperature heat (278–293 K) is to be removed from an outpost on the Moon designed to house a crew of six humans and operate for over 10 years continuously. Although the fiscal and socio-political circumstances have dampened the pace of research and development effort that would lead to the building of the lunar outpost, they are still an active part of the advanced concepts being pursued for space exploration.

Thermal control systems (TCS) are essential to maintain temperatures within acceptable levels in the various subsystems of any spacecraft or space-based facility. In most circumstances, the task is one of removing excess heat from different locations on the spacecraft and rejecting it into space. While simple, passive, thermal control techniques were adequate in most cases for spacecraft of the past, the complex and miniaturized spacecraft of the present and future pose difficult thermal design problems and involve sophisticated active control equipment. In all cases, the excess heat is rejected from the craft into space by radiative heat transfer, and a temperature differential of 40–50 K between the radiator and the environment is a practical minimum in order for the radiator size to remain reasonable. In most cases, a thermal sink that is significantly cooler than the radiator is readily available in space. The lack of availability of a low-temperature thermal sink makes the lunar base thermal cooling problem technically challenging.

The TCS ensures that the habitation modules for the crew, electronic and optical equipment, and other laboratories and instrumentation that are thermally conditioned are maintained within an acceptable temperature range. The variation of temperature on the lunar surface at the equatorial regions is shown in Fig. 1. Almost all desired locations for the frontier lunar bases are near the equatorial regions, and it can be seen that the surface temperature peaks at about 390 K during the 336-h lunar day. The direct dissipation of waste heat from the lunar base would be impractical given the adverse temperature gradient. This makes the rejection of moderate-temperature heat from the lunar base a challenging task.

Several techniques for moderate-temperature thermal control on lunar bases have been proposed, such as lunar soil thermal storage, shaded parabolic radiators, and heat pumps[2–5]. None of the techniques proposed is available off-the-shelf as space-qualified hardware. The effect of the lunar environment on the performance of the critical components of all the proposed systems is largely unknown. In the thermal storage concept, the excess heat from the base is pumped down, using a working fluid, to a heat exchanger buried deep under the lunar surface. The cool lunar soil provides the necessary low-temperature sink. The use of lunar soil as a thermal storage material would involve large-scale excavation deep into the surface of the Moon. In addition, the lack of knowledge about the high-temperature thermal properties of lunar soil deep under

†Paper IAF 94.0.4.486 presented at the *45th International Astronautical Congress*, Jerusalem, Israel, 9–14 October 1994.

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the surface is a prime reason for not pursuing this path presently. The concept of shaded light-weight radiators has been proposed. Shielding the radiator from the hot lunar soil could decrease the radiator operating temperature radically. The shade must be made of specular material and degradation of specular properties with time in the lunar environment is not known. Also, the fine dust kicked up from the surface around the base would be a serious concern since dust layers can radically impact the performance of the shaded radiators. At least for the frontier bases, a prudent approach would be to employ systems that rely on proven terrestrial technology and are least affected by exposure to the lunar environment. The concept of using heat pumps meets these criteria. In this concept, energy in the form of heat, or work, is supplied to the heat pump for removal of the heat from the low-temperature lunar base to the radiator at a much higher temperature. The temperature lift provided by the heat pump is sufficient for a reasonably sized radiator to dissipate the waste heat. The heat pump concept is also somewhat dependent on the degradation of the radiator surface properties with time, but much less so than the shielded radiator concept. It is this heat pump concept that will be pursued in this paper.

It should be noted that, while they are not commonplace, there are many space applications where a heat-pump-augmented thermal control system would be beneficial, besides a lunar base. All situations where a thermal sink with a 40–50 K temperature differential is not easily available would be candidates for the heat-pump-based TCS. Examples include lunar rovers that have long life expectancies and a Mars lander with high heat flux dissipation (the predominantly carbon dioxide atmosphere would make radiation difficult). There are some proposals for spacecraft to be operated in very low Earth orbits, and such craft would operate in a higher thermal sink environment. Furthermore, with the miniaturization of spacecraft subsystems and the push toward smaller spacecraft, the heat pump TCS would allow for compact radiators. The compact radiators would also prove attractive for military applications.

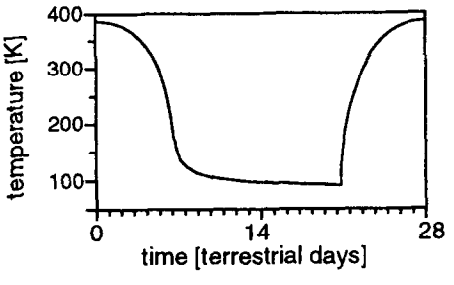


Fig. 1. Temperature variation of the surface at the lunar equator[1].

2. HEAT PUMP CONSIDERATIONS

Several different concepts for heat-pump-based TCS for the lunar base can be envisaged. It is possible to have a central thermal bus with heat pumps located at the radiator, or to have a distributed thermal bus with the heat pumps located at the load sources. In addition a combination of the two concepts is also possible. While both concepts have their inherent advantages and disadvantages, previous work has shown that a single centralized thermal bus concept makes the most sense from a mass standpoint[4]. Hence, the centralized concept will be considered in this paper.

Heat pumps have been in use for a long time for terrestrial applications for air conditioning and refrigeration. For large thermal loads, heat pumps operating on the Rankine, absorption, or reverse Brayton thermodynamic cycle have been used. The most common cycle for terrestrial applications is the Rankine cycle, also called the vapor compression cycle. It involves two isobaric and two adiabatic processes and requires shaft work. Absorption cycles differ from the Rankine cycle by requiring heat energy, not shaft work, to provide the temperature lift for the heat flux. Reverse Brayton cycle heat pumps utilize two isobaric and two isentropic processes in their cycle and employ a gas coolant rather than a liquid–vapor coolant. Stirling cycle heat pumps have been employed for cryogenic temperature ranges and low heat loads. In this paper, we consider the Rankine, absorption, and reverse Brayton heat pumps.

To optimize the mass of the heat-pump-based TCS, all promising options have to be evaluated and compared. This comparison requires the careful choice of system operating parameters, working fluids, and component masses. As with all comparative studies, a figure-of-merit must be established for the parametric analyses. This common yardstick will be the basis on which one concept will be favored over the others. For space applications, the total mass of a system is a good figure-of-merit. The single largest fraction of any space mission is the launch cost from Earth. The cost of launch from Earth to low Earth orbit (LEO) is in the neighborhood of $10,000 per kilogram and is much higher for transport to the lunar surface. For this optimization study, total mass of the TCS will be the objective function that will be minimized. To keep this study simple, the following assumptions are made: the systems are modeled for full-load operation, and the implications and power penalties at off-design and part-load conditions are not considered. However, the surface temperature of the lunar regolith varies from 80 to 390 K during the lunar day. This variation in the regolith temperature indicates that the temperature lift and the load of the heat pump would vary as a function of the time of day. For this reason, the performance of the heat pump at partial

conditions is important and will be studied in detail in the future. While evaluating system mass, the control components are not accounted for since the control system masses for the various concepts do not vary significantly and thus absolute mass is very small in comparison to the total TCS mass. Redundancy requirements are not considered here.

3. PROBLEM DEFINITION

Rankine, absorption, and reverse Brayton cycle heat pumps are considered for a lunar base moderate-temperature TCS. Analysis is performed to find the optimal system based on the least-total-mass criterion. The base, designed for four to eight crew members is presumed to have an electrical consumption of 100 kW. It is also assumed that this base is located in the equatorial region of the Moon. The cooling load and the acquisition temperature are described below.

3.1. Cooling load

Several previous studies[3,4,6] have estimated the electrical power input of a lunar base to be about 100 kW. To protect the crew and equipment from harsh radiation and from micrometeoroid impact, the walls are made of thick insulating material, and the net heat exchange with the outside environment through the walls is negligible. The metabolic heat generated by the crew members (~150 W/crew member) is negligible in comparison to that generated by the electrical input. It is assumed that the net thermal mass within the lunar base does not vary significantly with time and all the electrical energy is finally dissipated as heat. Based on these considerations, the total cooling load for the lunar base is fixed at 100 kW.

3.2. Acquisition temperature

The acquisition loop collects the excess heat from the base and transports it to the heat pump. Within the habitable regions of the base, for safety reasons, a chilled water loop will be used to acquire and transport the heat to the heat pump. Certain non-toxic trace additives may be added to water to depress its freezing point. For this study, it was decided that a single cooling loop operating at 275 K, the lower of the two Space Station cooling loop temperatures, would be used. The variation of the temperature of the coolant in the acquisition loop has to be chosen such that, for a reasonable mass flow rate of coolant, the given cooling load can be handled. With water as the coolant and a 5 K variation in the temperature, the mass of the coolant in the acquisition loop would be a reasonable 4.8 kg per second.

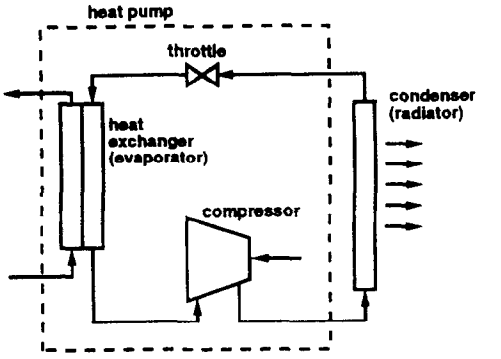


Fig. 2. Rankine cycle heat pump configuration, Case A.

4. RANKINE CYCLE HEAT PUMP

A study was conducted to optimize a heat pump operating on a Rankine cycle. Two different configurations were considered. In the first, termed Case A, the heat pump is directly connected to the rejection loop. In this configuration, the condenser of the heat pump and the radiator are one and the same, leading to mass savings. In this case, there is only one fluid flowing through the heat pump and the radiator. Figure 2 depicts a simple schematic of this configuration. Figure 3 is a schematic of Case B, where the condenser of the heat pump exchanges heat with a coolant loop connected to the radiator. In addition to adding extra components to the system, this configuration also imposes pressure and temperature drop penalties on the system. However, this configuration offers very attractive operational advantages. In Case B, it is relatively easy to connect several modular heat pumps in parallel without concern about proper flow distribution. The loss of refrigerant on one side of the condenser will not affect the other side, thereby enhancing the survivability of the overall system. In addition, repairability and incorporation of redundancy requirements are much improved. This concept may also be particularly valuable should the radiator and/or transport lines prove susceptible to micrometeoroid damage.

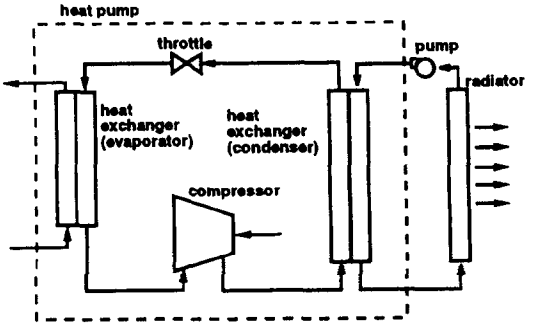


Fig. 3. Rankine cycle heat pump configuration, Case B.

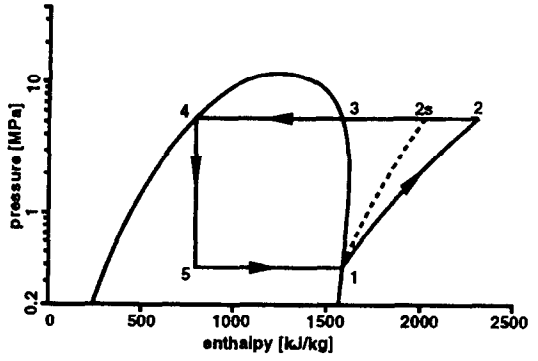


Fig. 4. Pressure–enthalpy diagram for a Rankine cycle heat pump.

4.1. *Analysis*

In order to develop an estimate of the mass of the heat pump TCS, the masses of each of the subsystems and plumbing were computed. Figure 4 is a pressure–enthalpy diagram for the Rankine cycle. A compressor pressurizes the working fluid from state 1 to 2. The working fluid is cooled in a condenser from state 2 to state 4, followed by expansion in a throttle valve to state 5. Low-pressure liquid vaporizes in an evaporator and attains state 1.

4.1.1. Compressor. The working fluid in the vapor state is compressed from state 1 to 2 by the compressor. Ideally, this process would be isentropic (1–2 S), but in reality it is non-isentropic (1–2) due to the irreversibilities. The properties of the refrigerants used in the modeling are real properties obtained from an inhouse computerized property database. Deviations from the ideal behavior in the compression occur due to mechanical, electrical (motor), and electronic (controller) inefficiencies, and fluid friction. In state-of-the-art aircraft compressors of comparable size these efficiencies are[7]: $\eta_{mech} = 0.95$, $\eta_{cont} = 0.91$, $\eta_{elec} = 0.94$, and $\eta_{fluid} = 0.75$. The overall efficiency of the compressor is $\eta_{total} = 61\%$. Since the compressor would be located outside the air-conditioned space, all the energy supplied to it during steady-state operation is used to compress the fluid, heat it, and radiate to the vacuum of space. It has been found that radiation losses from the compressor surfaces are negligible in comparison, so all the energy goes into compressing and heating the refrigerant. The mass of a state-of-the-art compressor for aircraft applications is 0.45 kg/kW of cooling load. In our application, the cooling load will remain the same (100 kW) during the mass optimization, but the rejection heat load will vary depending on the temperature lift. An analysis showed that the compressor specific mass was 0.202 kg/kW of the rejected heat[8].

4.1.2. Condenser. Between states 2 and 3, the compressed superheated vapor cools to saturated conditions, and between 3 and 4 it condenses to saturated liquid. Ideally, this process would be isobaric, but due to pipe friction and configuration changes in the condenser (such as elbows and bends), a small pressure drop will occur. In Case A, processes 2–4 occur in the radiator, since the radiator and condenser are one and the same. The mass of the radiator and the mass of the plumbing from compressor (heat pump) to the radiator will be discussed later. In Case B, the condenser is a heat exchanger separating the heat pump from the rejection loop. A specific mass of 2.72 kg/kW of heat transferred is assumed for this two-phase heat exchanger[4]. The temperature difference across the two flows in the condenser is set to 5 K, and a pressure drop that is equivalent to a 1 K temperature drop in each flow loop through the condenser is assumed.

4.1.3. Throttle valve and evaporator. Between states 4 and 5, the refrigerant is throttled. The process is assumed to be adiabatic. The mass of a throttle valve is negligible in comparison to the masses of the other components of the heat pump. From state 5 to state 1, the refrigerant absorbs heat from the coolant circulating in the acquisition loop of the lunar base. The absorption of the cooling load takes place in the evaporator. This process, again, deviates from the ideal isobaric process due to a frictional pressure drop in the condenser (set here to be equivalent to a 1 K temperature drop). The temperature differential between the coolant in the acquisition loop and refrigerant in the evaporator is set at 5 K. The specific mass of the evaporator is taken to be 2.72 kg/kW of cooling load.

4.1.4. Rejection loop. In Case A, the plumbing from the compressor (heat pump) to the radiator forms the rejection loop, hence the rejection loop coolant and the heat pump refrigerant are one and the same. Here, the length of the loop is determined by the height of the radiators and the distance from the heat pump to the radiators[8]. The total pressure drop in the rejection loop is taken to be the equivalent of a 1 K temperature drop, based on the guidelines set for good design practice. Allowing a large pressure drop in the rejection loop would deteriorate the overall performance of the cycle. The size of the piping that would permit the specified pressure drop is calculated, and its thickness is set to be the larger of 0.5 mm or the hoop-stress-dictated value with a factor of safety of three. It is assumed that the piping is made of a light-weight, high-strength aluminum alloy with suitable insulation.

In Case B, the rejection loop couples the condenser of the heat pump to the radiator and has a mechanical pump to circulate the coolant in the rejection loop. The mass of the pump and its power penalty would have to be taken into account in the mass estimation and optimization. The mass and power requirements are computed using the following expressions given by Dexter and Haskin[9]:

$$W_{pump} = \frac{\dot{m}H}{3600 \cdot 550 \cdot \eta}$$

$$M_{pump} = 5.61 \cdot \left(\frac{\dot{m}}{60 \cdot \rho}\right)^{0.75}$$

where W_{pump} is the pump power in horse-power, M_{pump} is the mass of the pump in pounds, $\dot{m}$ the mass flow rate in pounds per hour, H the pump head in feet, ρ the fluid density in pounds per cubic foot, and η the pump efficiency assumed to be 0.25. The criteria used for pipe thickness, total pressure drop, and pipe mass calculation were the same as in Case A. The coolant used in the rejection loop was ammonia because of its superior heat transport characteristics[8].

4.1.5. Radiator. The function of the radiator is to reject the waste heat from the base. The heat rejected by the radiator is given by

$$Q = A\epsilon\eta\sigma(T_{reject}^4 - T_{sink}^4)$$

where A is the radiator area, ϵ is the emissivity, η the fin efficiency, σ the Stefan–Boltzman constant, and T_{reject} and T_{sink} the radiator and sink temperatures, respectively. The estimated peak sink temperature for a vertically mounted radiator at the lunar base[3] is 321 K. Most sources point to an ϵ of 0.8 and η of 0.7 for such radiators. Several widely differing values are reported in the literature for specific masses for radiators[3,4,10,11]. A specific mass of 2.5 kg/m^2 is assumed in the present analysis for vertical two-sided radiators.

4.1.6. Power supply. The Rankine cycle requires electric power to provide the shaft work required by the compressor. The capacity of the power station on the lunar base would have to be increased in order to provide the necessary power. A review of the literature shows that there is no consensus on the mass penalty associated with the power supplies[3,4,12,13]. The values quoted for the SP 100 class nuclear power sources are about 30 kg/kW of electric power. Although the SP 100 program is no longer active, power sources with penalties of the order of 30 kg/kW of electric power would have to be developed to make missions such as lunar habitats feasible. For this reason, a 30 kg/kW power penalty is chosen in this study.

4.1.7. Refrigerants. One of the important issues is the choice of refrigerant for the heat pump. The commonly used vapor compression cycle refrigerants for terrestrial applications are R11, R12, R113, R114, and R717. R113 and R114 were eliminated from this list because of potential for condensation of the vapor in the compressor. Due to a low critical temperature (385 K) and low COP, R12 was not selected. R11 and R717 are used as potential candidates. Using real properties for the refrigerants, the COP of the cycle is computed for a range of rejection temperatures and plotted in Fig. 5.

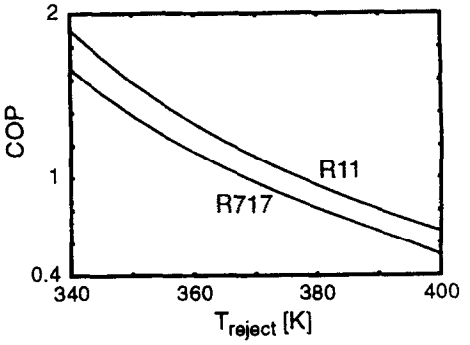


Fig. 5. COP as a function of rejection temperature for R11 and R717.

4.2. Results

Values for COP and the mass of the piping were computed and tabulated for varying rejection temperatures using the mass models presented above. These values were imported into a spreadsheet and linearly interpolated where necessary. The mass optimization was performed using a spreadsheet. The radiator temperature, or equivalently the heat pump temperature lift, was varied and the masses of the various components were computed. For Case A, the analyses were performed for two refrigerants, R11 and R717. The results are shown in Fig. 6. For Case B, R11 or R717 were used as the heat pump working fluid and R717 was used as rejection loop coolant. The results are presented in Fig. 7. When R11 is used as the working fluid for the heat pump, the optimal TCS mass is 6108 kg at a radiator temperature of 371 K for Case A. For Case B, the optimal TCS mass is 5940 kg at a radiator temperature of 362 K. The radiator mass is higher in Case B than in Case A due to the lower radiator temperature. In addition, the condenser mass and rejection loop pump mass add to the total TCS mass in Case B. However, the superior transport properties of R717 in the rejection loop reduce its mass significantly in Case B and result in the overall TCS mass being lower for Case B than Case A when R11 is used as the refrigerant. When R717 is used as the working fluid in the heat pump, the optimal mass of the TCS is 5515 kg at a radiator temperature of 362 K for Case A. For Case B, the corresponding values are 6392 kg and 360 K, respectively. Due to the lower radiator temperature, additional heat exchanger, and rejection loop pump,

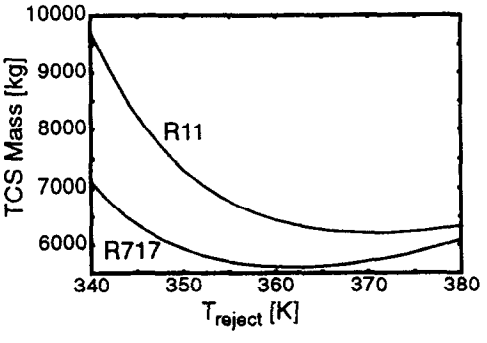


Fig. 6. TCS mass as a function of rejection temperature for Case A.

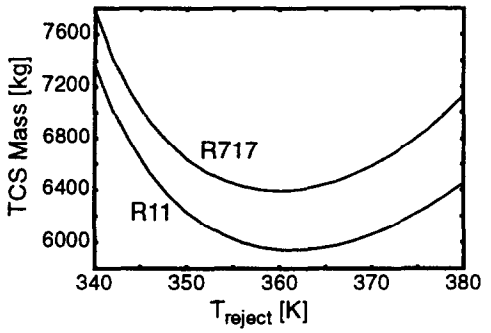


Fig. 7. TCS mass as a function of rejection temperature for Case B.

Case B is more massive than Case A. Among the cases considered, R717 with the Case A configuration has the lowest TCS mass, 5515 kg, followed by 5940 kg for Case B with R11 as the heat pump fluid and R717 as the rejection loop coolant. For the additional 425 kg, the latter system offers significant operational and safety improvements and would be the system of choice for Rankine cycle systems.

5. HEAT-DRIVEN HEAT PUMPS

The Rankine cycle heat pump discussed in the previous section requires energy in the form of shaft work to provide the temperature lift. Heat pumps that require shaft work to operate fall in the class of work-driven heat pumps (WDHP). Another class of heat pumps, the heat-driven heat pumps (HDHP), operate by using heat to provide the necessary temperature lift for the heat pump. The most popular HDHP that has seen terrestrial use is the absorption heat pump. Waste heat from power sources or process heat from chemical reactors is used to drive the cycle. In the lunar base application, the waste heat from a nuclear power source would be a potential source of the heat. Should this source not be available, a dedicated solar energy collector system would be used. A solar concentrator for heat collection has appeal since it is self-adaptive in that the availability and collection profile of solar energy would be coincident with the thermal energy needs of the heat pump. During the lunar noon, when the effective sink temperature is at its peak and the demand for temperature lift from the heat pump is a maximum, the energy available from the solar collector will also be at its maximum.

Figure 8 is a simplified schematic of an absorption cycle heat pump. The condenser where the "temperature-lifted" heat is transferred to the radiator for rejection (states 1–2), the throttling to lower pressure (states 2–3), and the intake of the moderate-temperature heat from the acquisition loop to the heat pump through the evaporator (states 3–4) are all similar to the Rankine cycle described above.

The point of departure from the Rankine cycle occurs in the process of compression of the low-pressure vapor to higher pressures. The power-intensive vapor compressor of the Rankine cycle is replaced by an absorber and generator system. The refrigerant goes into solution with a carrier fluid in the absorber (states 4–5), and this liquid solution is pumped to a higher pressure by a small liquid pump (states 5–6). The addition of heat to the solution of refrigerant and carrier fluid at high pressure in the generator causes the separation of the refrigerant at high pressure (states 6–7). The solution that is weak in refrigerant is circulated back into the absorber by a small pump (states 7–8). The attractiveness of this cycle for terrestrial applications is replacement of the compressor work in the Rankine cycle with readily available waste heat from other processes and negligible mechanical work.

In the lunar base scenario, heat must be rejected from both the condenser and absorber. The heat rejected at the absorber and the condenser will be at a lower temperature than the source heat at the generator, thereby requiring a large radiator area. The magnitude of the mass penalty for such a system would depend on the sink, radiator, and lunar base temperatures. As stated before, a refrigerant and a carrier fluid are required for the absorption heat pumps. The two most common systems, ammonia–water, and water–lithium bromide systems will be considered in this study.

5.1. *Ammonia–water heat pump*

In the ammonia–water system ammonia is the working fluid and water is the carrier fluid. Even the simplest setup of an ammonia–water heat pump is more complex than the basic absorption heat pump presented in Fig. 8. One of the major reasons for the added complexity arises from issues related to the presence of small amounts of water in the high-pressure ammonia released from the aqueous solution in the generator. A small amount of water can cause a considerable drop in the condenser temperature and make the cycle ineffective: for example, a 0.5 mass percent of water can cause a 10 K temperature drop. Dephlegmators (heat exchangers that use cooling coils to condense the water from the

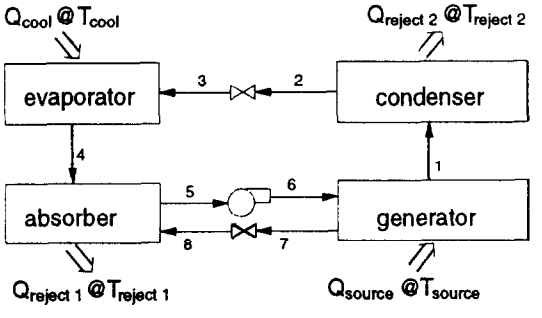


Fig. 8. Simplified schematic of an absorption heat pump.

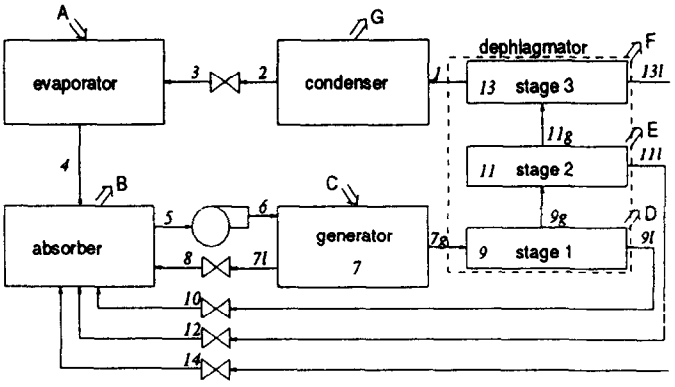


Fig. 9. Simple ammonia–water heat pump.

mixture) are used to rid the ammonia vapor of water. Figure 9 is a simple schematic of the ammonia–water heat pump. From analysis, it was found that a three-stage dephlegmator was needed for this application. An enthalpy-concentration diagram of the thermodynamic processes is shown in Fig. 10. The dashed lines denote the two-phase region at the low-pressure level, and the dash–dotted lines denote the high-pressure saturation lines. Constant temperature lines at the high-pressure levels are denoted by dotted lines. The solid lines in the diagram denote the processes in the cycles and the state points correspond to the ones marked in Fig. 9. A *g* or *l* suffixed to a state point corresponds to the states of the vapor and liquid phases, respectively, of the two-phase mixture[14]. In order to attain a 99.9 mass percent ammonia concentration in the refrigerant part of the loop a three-stage dephlegmator was necessary as ascertained by the thermodynamic analysis. Using the property values provided in the ASHRAE handbook[15] for ammonia–water mixtures, the coefficient of performance (COP) of the cycle was found to be 0.0344. In order to improve the overall performance of the system, a more complex system with better thermal management was considered. A waste heat recovery heat exchanger was used between the absorber and generator for this purpose, as shown in Fig. 11. Though the COP of the system improved, it was still only 0.0998. Due to the very low COP of this system, it was not pursued further.

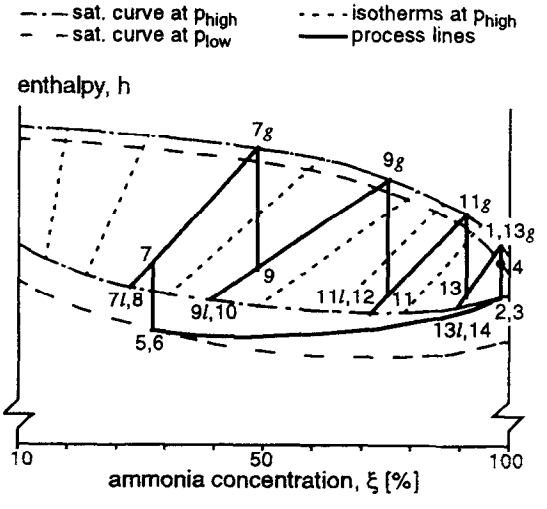


Fig. 10. Concentration–enthalpy diagram of a simple ammonia–water heat pump.

5.2. *Water–lithium bromide heat pump*

The water–lithium bromide system is a popular HDHP system for terrestrial systems due to the relative ease with which the refrigerant (water) can be separated from the carrier fluid (lithium bromide). Figure 12 depicts a simple schematic of a water–lithium bromide system. Here, superheated steam at high pressure (state 1) leaves the generator and condenses to saturated liquid in the condenser (state 2). The liquid is subcooled in the precooler (state 3) and expands in the valve to form a low-pressure liquid–vapor mixture (state 4). This mixture absorbs heat from the cooling load to form a low-pressure saturated vapor in the evaporator (state 5). The vapor absorbs heat in the precooler to form supersaturated vapor (state 6) before entering the absorber. In the absorber, the supersaturated steam mixes with the strong solution of the mixture. The heats of condensation and solution are removed from the

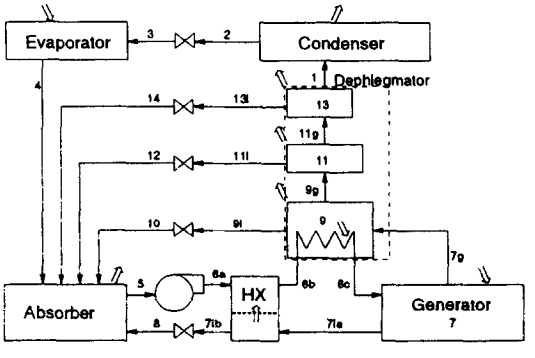


Fig. 11. Ammonia–water heat pump with internal heat recuperation.

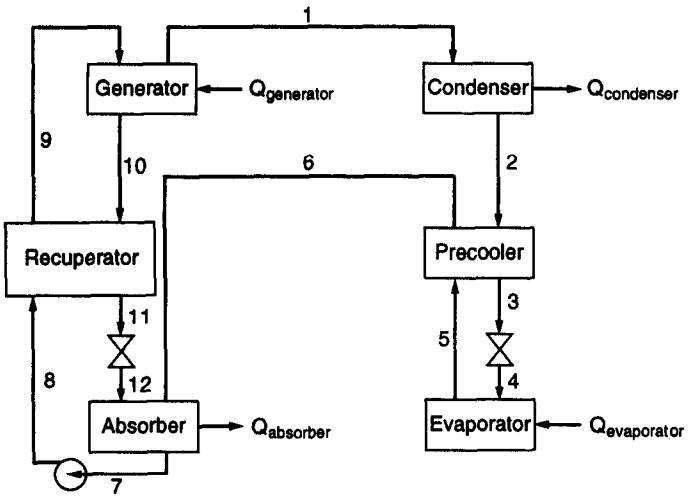


Fig. 12. Schematic of a water–lithium bromide heat pump.

absorber by cooling coils. The low-pressure solution is pumped to a higher pressure, and it absorbs some heat in the recuperator. In the generator, the addition of heat separates pure steam from the weak solution. The strong solution from the generator (state 10) reenters the absorber to complete the cycle, after rejecting some heat in the recuperator.

This thermodynamic cycle has four degrees of freedom for a specified heat load, namely, cooling load temperature, generator temperature, and concentrations of the weak and strong solutions. The cooling load temperature has already been picked based on the lunar base requirements. Based on the onset of crystallization, the upper bound of the strong solution concentration can be set. Under these conditions COP is a function of generator temperature and weak solution concentration. It would be desirable, and elegant, if the mass optimization procedure for this cycle, analogous to the Rankine cycle, is one where the COP can be expressed as a function of T_{high} alone for a given T_{cool}. The following procedure is adapted in order to attain such a relationship.

Figure 13 shows the variation of COP with ξ_{weak} for generator temperatures of 500, 600, and 700 K. It can be seen that the lower the concentration, the better the COP. Hence, for better performance, it is desirable to operate the cycle with the weak solution concentration as low as permissible. The variation of T_{high} with ξ_{weak} for generator temperatures of 500, 600, and 700 K is shown in Fig. 14. It can be seen that there is a one-to-one correspondence between ξ_{weak} and T_{high} in the region of $\xi_{weak} \leqslant 60\%$. From these two observations, it is possible to express ξ_{weak} as a function of T_{high} for a given T_{cool} and T_{gen}. So COP can be specified in terms of T_{high} and T_{gen}. Figure 15 is a plot of the COP as a function of T_{gen} for T_{high} in the range of 360–400 K. It can be seen that, for any given T_{high}, there is a distinct maximum for COP. Assuming that the operation of a heat pump at these maxima is feasible, it is found that these COPs are linearly dependent on T_{high}, as seen in Fig. 16. Using this

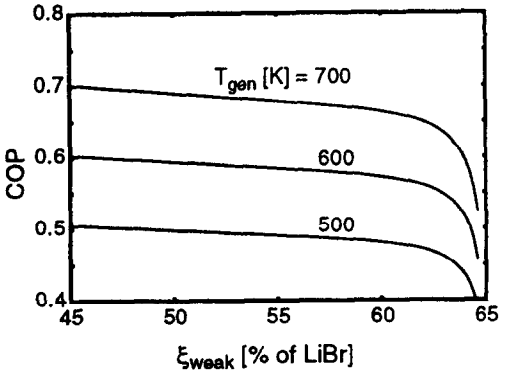


Fig. 13. Variation of COP with ξ_{weak}.

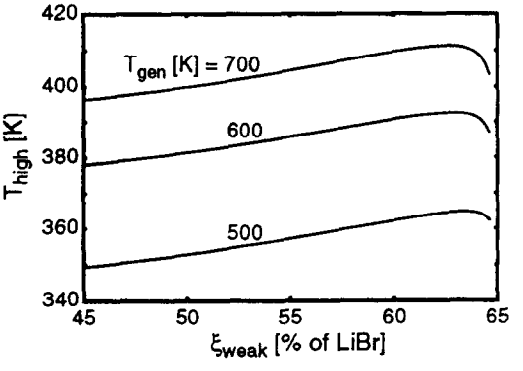


Fig. 14. Variation of T_{high} with ξ_{weak}.

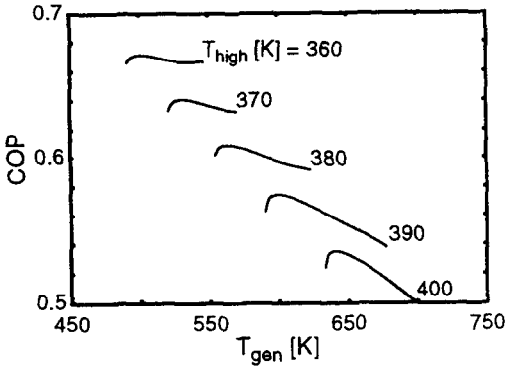


Fig. 15. Variation of COP as a function of T_{gen} for several T_{high}.

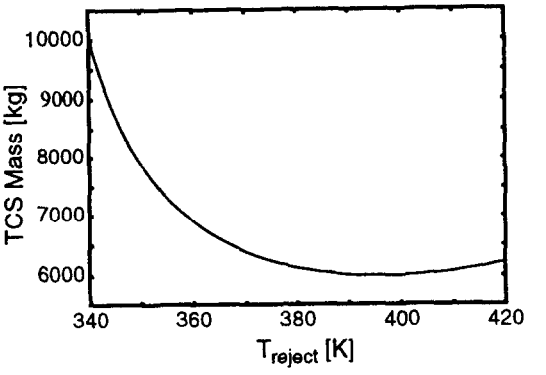


Fig. 17. TCS mass of a lithium bromide–water heat pump system.

relationship, the radiator temperature and the optimal mass of the TCS are determined.

It is to be noted that since water is used as a refrigerant in this system, a T_{cool} of 280 K, not the 270 K used for the Rankine cycle, is used here. This increased acquisition temperature may not be acceptable for certain sensor cooling needs on the lunar base. Nevertheless, the analysis was performed in order to study mass estimates. The specific masses for the various absorption cycle components are not readily available. However, the major components of the heat pump (evaporator, condenser, absorber, generator, precooler, and recuperator) can all be approximated as heat exchangers, and their masses estimated based on the rate of heat transfer occurring inside them. Such an approximation is recognized to be simplistic, and it will underpredict actual masses. For this reason, such an analysis is termed an "optimistic mass estimate."

The heat loads for all absorption heat pump components are then estimated to predict the total mass of the heat pump. Detailed cycle analysis has shown that the total heat transferred in all components can be expressed in terms of the cooling load and the COP. This results in an expression for the absorption heat pump mass, M_{HP}, of

$$M_{HP} = (2.72\ \text{kg/kW}) \cdot Q_{total}$$

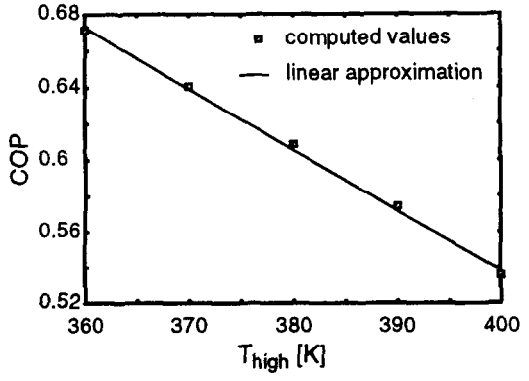


Fig. 16. Variation of COP with T_{high}.

where

$$Q_{total} = Q_{cool}\left(2 + \frac{\frac{Q_{int}}{Q_{gen}}}{COP}\right)$$

Q_{cool} is the cooling load, Q_{int} is the internal heat load (sum of heat loads in recuperator and precooler), and Q_{gen} is the heat supplied to the generator. COP as a function of T_{high} is obtained from the linear approximation shown in Fig. 16. The ratio Q_{int}/Q_{gen} can be assumed constant for the range of operation of interest for the heat pump.

In addition to the mass of the rejection loop and the radiators, whose estimates are similar to those presented for the Rankine cycle Case B, an additional transport loop mass has to be considered for the absorption cycle. This transport loop carries the energy from the high-temperature heat source to the heat pump. The mass of this transport loop includes the tubes, transport fluid, pump, and the power penalty for the pump. The length of the piping is assumed to be 500 m, and the tube diameter is optimized for a minimum mass consistent with a pressure drop of no more than 10%. The transport fluid of choice for the 525–975 K operating range is water. A detailed analysis was performed[14] to show that a linear mass model can be developed for the working range

$$M_{trans.loop} = (2.9\ \text{kg/kW}) \cdot Q_{source}$$

5.3. *Results*

The mass models were incorporated into a spreadsheet for total mass optimization. Figure 17 depicts the mass of the lithium bromide–water TCS as a function of radiator temperature. It can be seen that the optimum is at a rejection temperature of 396 K and the corresponding TCS mass of about 6000 kg. The following assumptions were made and each of these underpredicts the total mass: (1) the components were approximated as heat exchangers; (2) pressure drops in the heat exchanger components were neglected; (3) no mass penalty was assigned for

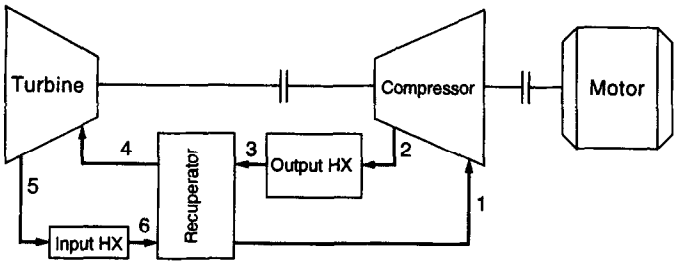


Fig. 18. Schematic of reverse Brayton cycle with recuperation.

the high-temperature source; and (4) T_{cool} was taken to be 280 K, rather than 270 K. For these reasons it is concluded that the optimal Rankine cycle described earlier has a better figure-of-merit than the optimal water–lithium bromide cycle described here.

6. REVERSE BRAYTON CYCLE HEAT PUMP

This section describes the analysis of a TCS concept based on the reverse Brayton cycle heat pump. Overall mass and performance data have been developed for this concept. The comparison of the optimized TCS mass is primarily with the Rankine cycle heat pump. The philosophy used is as follows: an ideal cycle with optimistic mass models was chosen for the reverse Brayton cycle. If the results of this idealized model proved to be promising in comparison with the reference case, refinements to the ideal model would be warranted. If the overall mass of this idealized system was larger, then the reference case would be preferred over the reverse Brayton cycle option.

6.1. Thermodynamic and mass model

The ideal Brayton cycle consists of two isobaric and two isentropic processes. It is realized with a compressor, turbine, and two heat exchangers. It is similar to the Rankine cycle except that (1) the throttling is replaced by an isentropic expansion in a turbine and (2) the working fluid always remains gaseous. These differences point to the inherent advantages and disadvantages of the reverse Brayton cycle. The energy of expansion is not wasted in a throttling device. Instead, it is recovered in the turbine, thereby reducing the amount of shaft work required. Also, there exists the possibility of using a combined refrigeration and power cycle, which would simultaneously utilize two turbines and one compressor and effectively utilize waste heat that may be available from the power supply subsystem. Since the working fluid remains gaseous throughout the cycle, the mass and volume flow rates are large. This is because the sensible heats are small on a unit mass basis compared to the latent heats associated with liquid–vapor phase change. The isobaric heat exchanger is not isothermal and this not only causes excessively high compressor and low turbine outlet temperatures, but also decreases cycle performance.

Under most conditions, it is advantageous to modify the reverse Brayton cycle, as described above, with internal heat transfer for heat recuperation. This can be especially useful if large temperature lifts with relatively low pressure ratios are desired. Figures 18 and 19 depict the schematic and a temperature–entropy representation, respectively, of a reverse Brayton cycle heat pump with recuperation. An ideal recuperator, $T_4 = T_6$, is assumed.

The thermodynamic model takes into account the inefficiencies of the components. In real equipment, the compression and expansion cannot be isentropic and heat transfer will not be isobaric. The efficiency of the compressor is defined by

$$\eta_c = \frac{w_{\text{isentropic}}}{w_{\text{real}}}$$

The analogous expression for the turbine is

$$\eta_t = \frac{w_{\text{real}}}{w_{\text{isentropic}}}$$

where $w_{\text{isentropic}}$ is the isentropic work supplied or generated and w_{real} is the actual work supplied or generated.

The heat exchangers will be modeled with a relative pressure drop:

$$\eta_{HX} = \frac{p_{\text{out}} - p_{\text{in}}}{p_{\text{in}}}$$

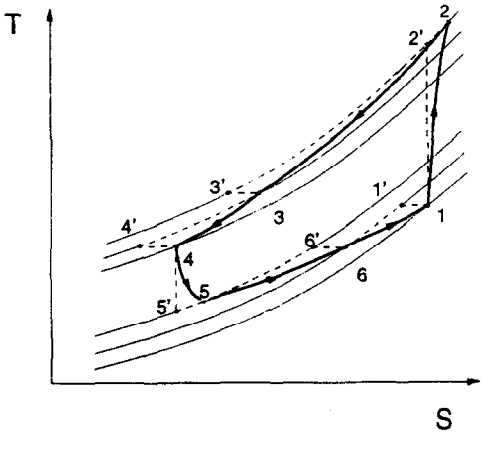


Fig. 19. T–s representation of reverse Brayton cycle with recuperation.

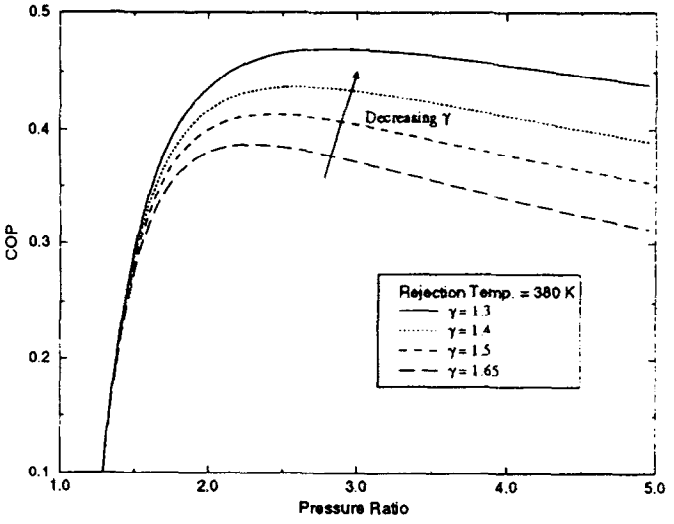


Fig. 20. COP vs ϕ for several values of γ for recuperated cycle.

where p_{in} is the inlet pressure and p_{out} is the outlet pressure.

The inefficiencies represent a trade-off between size and mass on one hand and performance on the other. For this study, values of $\eta_c = 0.8$ and $\eta_t = 0.87$ are assumed. These are the best achievable efficiencies with equipment sizes at the upper limit of the acceptable range. It can be shown that the overall COP is

$$\mathrm{COP}(\Phi) = \frac{T_6\left[1 - (\xi_{HX}\Phi)^{\frac{1-\gamma}{\gamma}}\right]\eta_t}{\frac{T_3}{\eta_c}\left[\phi^{\frac{\gamma-1}{\gamma}} - 1\right] + \eta_t T_6\left[(\xi_{HX}\Phi)^{\frac{1-\gamma}{\gamma}} - 1\right]}$$

where:

$$\Phi = p_2/p_1$$

$$T_6 = T_{cool}$$

$$T_3 = T_{reject}$$

$$\xi_{HX} = (1 - \eta_{HXin})\,(1 - \eta_{HXout})\,(1 - \eta_{recin})\,(1 - \eta_{recout})$$

and γ is the specific heat ratio of the fluid. The recuperators are modelled in a manner similar to that for the heat exchanger with a relative pressure drop:

$$\eta_{rec} = (p_{in} - p_{out})/p_{in}$$

The 10% total pressure drop is assumed to be split between the heat exchanger and recuperator: $\eta_{regin} = 0.95$, $\eta_{regout} = 0.95$, $\eta_{HXin} = 0.95$, $\eta_{HXout} = 0.95$. The COP is plotted as a function of ϕ in Fig. 20 for several values of γ.

In this case, again, COP is obtained as a function of condenser temperature alone using the following procedure: the COP for a given working fluid and cooling temperature, T_1, is a function of both the rejection temperature and the pressure ratio. But, for a mass optimization, the COP is to be obtained as a function of the condenser temperature only. From Fig. 21, it can be seen that, corresponding to every rejection temperature, there exists a maximum COP, occurring at a particular pressure ratio, which keeps shifting to the right with increasing rejection temperature. For a given rejection temperature, the maximum value of the COP is chosen and the pressure ratio corresponding to this COP is taken to be the operating pressure ratio. This procedure ensures optimization of COP with respect to pressure ratio, thereby reducing the COP to a function of one independent variable. This procedure was implemented using a Fortran77 code, and the results are plotted in Fig. 22 for pure helium. The COP varies almost linearly with rejection temperature. No interpolation is performed to obtain COP as a function of radiator temperature.

The pressure ratio that corresponds to the maximum COP for all condenser temperatures is 2.3:1 when pure helium is used as a working fluid. This is shown in Fig. 21. It should also be noted that the maximum COP for the recuperated cycle occurs at a single constant pressure ratio, irrespective of the condenser temperature. The COP drops almost linearly with condenser temperature.

6.2. *Selection of working fluid*

The desirable properties for the working fluid can be deduced from the results of the previous analysis. A description of the working fluid properties follows.

6.2.1. Specific heat ratio. Lower specific heat ratios (γ) permit higher coefficients of performance. So the most obvious decision is to choose a fluid with a low γ. Carbon dioxide ($\gamma = 1.3$) and ethylene ($\gamma = 1.2$) are well suited, based on this criterion. An argument can be made that these two fluids do not make suitable working fluids because of condensation problems, but the operating pressure can be manipulated such that they are always in the gaseous

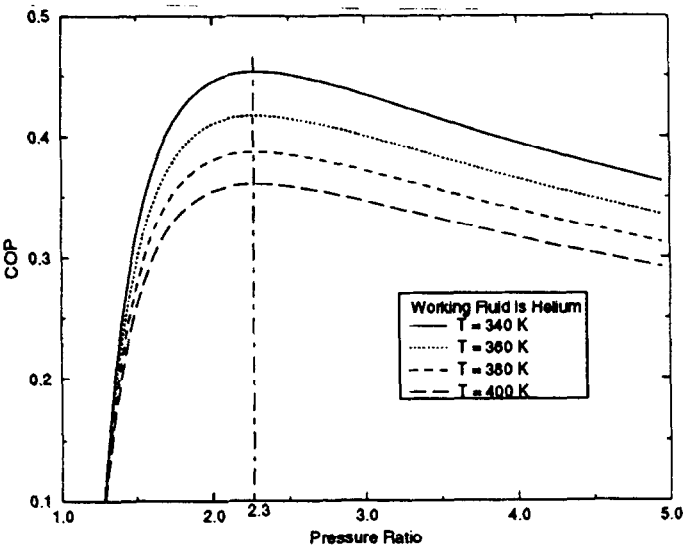


Fig. 21. COP vs ϕ for several rejection temperatures for the recuperated Brayton cycle.

phase. However, helium ($\gamma = 1.65$) and air ($\gamma = 1.4$) have also been considered in the analysis because the technology for using these working fluids has a proven track record.

6.2.2. Molecular weight. Low molecular weight gases have some advantage because of lower friction[16]. The molecular weight of the working fluid also affects the mass of the process equipment because of decreased fluid momentum.

6.2.3. Specific heat. Working fluids with low specific heats require a larger heat exchanger. A mixture of helium and xenon having a higher molecular weight is preferred because of the reduced mass of the heat exchanger and the recuperator[16]. Air, though a good working fluid, does not have good transport properties.

6.3. Mass optimization

The following mass models were utilized in order to estimate the mass of the TCS. The radiator, the power supply, and the rejection loop mass models have been dealt with in previous sections and are not repeated here.

6.3.1. Heat exchanger. It conveys heat between the heat pump loop and the rejection loop. The specific mass of the heat exchanger is 2.72 kg/kW of heat exchanged.

6.3.2. Recuperator. It exchanges heat between two gaseous fluids. Hence, the area of the heat exchanger and its mass are large. The specific mass of the recuperator has been estimated at 10 kg/kW of heat exchanged[17]. This is at a pressure drop of 5% across the recuperator. The recuperator mass also

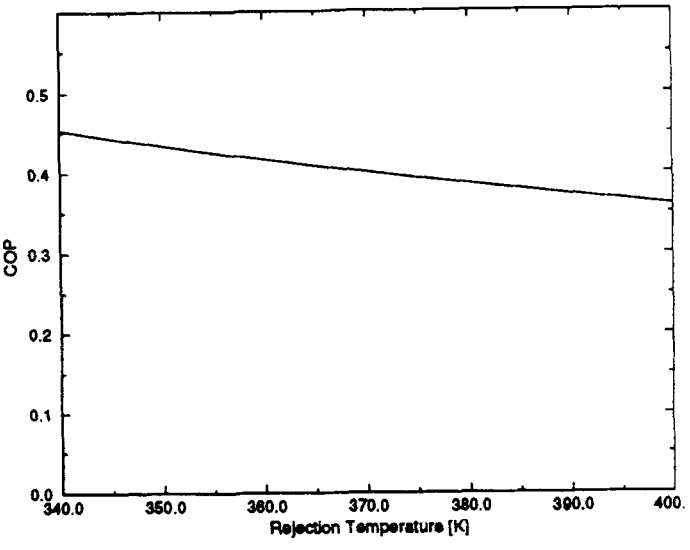


Fig. 22. COP as a function of rejection temperature only for recuperated Brayton cycle.

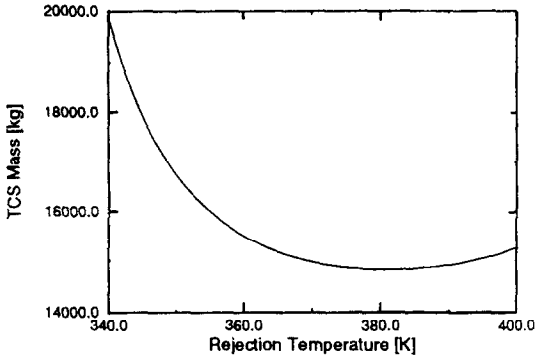


Fig. 23. TCS mass as a function of rejection temperatures for helium.

depends on the molecular weight and specific heat capacity of the working fluid. Notwithstanding, the recuperator specific mass is considered to be independent of the nature of the fluid for this simple analysis.

6.3.3. Rotating machinery. It consists of the turbine compressor and the motor mounted on a single shaft. A specific mass of 1 kg/kW of the power driving the turbine shaft is used in this analysis. The efficiencies anticipated are $\eta_t = 0.87$ and $\eta_c = 0.8$[17]. This empirical formula is valid in the range of 100–200 kW shaft power. Again, the turbine specific mass and performance (in terms of its isentropic efficiency) depend on the molecular weight of the working fluid. If the molecular weight is lower, less momentum is transferred.

6.4. Results

A mass analysis for the recuperated Brayton cycle TCS has been carried out for air, helium, and a fluid with $\gamma = 1.2$. This specific value ($\gamma = 1.2$) has been chosen to demonstrate how a fluid with a very low specific heat ratio would perform, disregarding all potential operational difficulties with such a medium.

Figure 23 shows the effect of rejection temperature on the TCS mass for helium. The TCS mass exhibits a minimum at a single rejection temperature.

Figure 24 shows results for the three different working fluids at an optimal rejection temperature and compares against the optimal Rankine cycle case. The power supply is by far the largest single mass fraction, followed by the radiator and the recuperator.

7. CONCLUSIONS

It has been shown that a heat pump based TCS would be the concept that is most suitable for moderate temperature heat rejection on lunar bases until other options, such as lunar soil thermal storage and shaded radiators, are well explored for their performance in the lunar environment. Rankine, absorption, and reverse Brayton cycle systems were analysed as candidate cycles for the heat pump. A thorough mass analysis using real properties of refrigerants and real mass models and performances was performed for the Rankine cycle. The mass of a Rankine cycle heat pump based TCS that used R11 as the heat pump fluid and R717 as the rejection loop fluid was identified as the system of choice for this cycle. Such a cycle rejected 100 kW of lunar base heat at 270 K to the lunar environment by raising the effective radiator temperature to 362 K. The total mass of this TCS was 5940 kg. This case was used as the benchmark.

Ammonia–water and water–lithium bromide absorption heat pumps were then analysed. For the present application, the COP of potential ammonia–water heat pumps were very low, in the neighborhood of 0.1, and so were not considered further. Due to the non-availability of accurate mass models for the components the water–lithium bromide heat pump, an underpredicted total TCS mass was obtained for

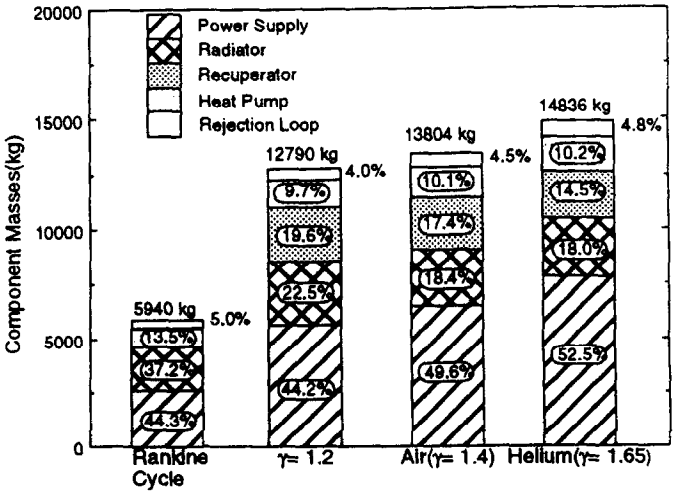


Fig. 24. Overall TCS and component masses for various working fluids.

this system using simplifications and ideal processes with no irreversibilities. It was argued that if this optimistic mass estimate were not significantly lower than the reference benchmark, a more detailed analysis would not be warranted. This was indeed the case, hence, this option was not pursued further.

A reverse Brayton cycle heat pump TCS was considered next and the mass of the optimal system was found to be more than twice that of the reference benchmark, 12,970 kg at a radiator temperature of 374 K.

The figure-of-merit for the TCS comparison was established to be the least mass of the TCS. Based on this criterion it is concluded that the Rankine cycle heat pump described above as the benchmark was the optimal solution for this application.

Acknowledgements—This research was supported by the NASA Goddard Space Flight Center under Grant NAG5-1572. The authors would like to extend their thanks to Mr Theodore D. Swanson of GSFC for his many helpful suggestions and discussions. We are grateful to Mr Ashok Nanjundan for his help with the reverse Brayton cycle analysis and Ms Connie Spencer for editing the manuscript.

REFERENCES

1. C. J. Cremers, R. C. Birkebak and J. E. White, Lunar surface temperatures from Apollo 12. *The Moon* **3**, 346–351 (1971).
2. L. C. Simonson, M. J. DeBarro, J. T. Farmer and C. C. Thomas, Conceptual design of a lunar base thermal control system. *Symposium on Lunar Bases and Space Activities in the 21st Century*, Houston, TX, April (1988).
3. M. K. Ewert, P. A. Petete and J. Dzenitis, Active thermal control systems for lunar and martian exploration. *20th Intersociety Conference on Environmental Systems*, Williamsburg, VA, July (1990).
4. T. D. Swanson, R. Rademacher, F. A. Costello, J. S. Moore and D. R. Mengers, Low-temperature thermal control for a lunar base. *20th Intersociety Conference on Environmental Systems*, Williamsburg, VA, July (1990).
5. F. A. Costello and T. D. Swanson, Lunar radiators with specular reflectors. *AIAA/ASME Thermophysics and Heat Transfer Conference*, Seattle, Washington, June ASME Publication HTD-Vol. **135**, pp. 145–150 (1990).
6. R. D. Waldron, Lunar base power requirements, options and growth. *Proceedings of Space* 90, pp. 1288–1297, Washington, D.C., AIAA (1990).
7. R. Murray, AiResearch, Los Angeles, CA; private communication (1991).
8. M. Gottmann and K. R. Sridhar, Thermal control systems for low-temperature heat rejection on a lunar base. *28th National Heat Transfer Conference*, pp. 148–157, San Diego, AICHE (1992).
9. P. F. Dexter and W. L. Haskin, Analysis of heat pump augmented systems for spacecraft thermal control. *19th AIAA Thermophysics Conference*, Snowmass, CO, June (1984).
10. L. Guerra, A commonality assessment of lunar surface habitation. *Proceedings of Space* 88, pp. 274–288, Washington, D.C. (1988).
11. B. Drolen, Heat pump augmented radiator for high power spacecraft thermal control. *27th Aerospace Science Meeting*, Reno, NV, January, Paper No. AIAA 89-0077 (1989).
12. G. A. Landis, S. G. Bailey, D. J. Brinker and D. J. Flood, Photovoltaic power for a lunar base. *Acta Astronautica*. **22**, 197–203 (1990).
13. E. J. Roschke and L. C. Wen, Preliminary system definition study for solar thermal dynamic space power systems. *JPL Technical Report D-4286*, JPL, Pasadena, CA, June (1987).
14. K. R. Sridhar and M. Gottmann, Thermal control systems for low-temperature heat rejection on a lunar base. *Final Report Grant NAG5-1572*, NASA/GSFC, October (1992).
15. *ASHRAE Handbook* 1989 *Fundamentals*. ASHRAE, New York (1989).
16. B. D. Wood, *Application of Thermodynamics*, second edition. Waveland Press, Prospect Heights, Illinois (1991).
17. A. Juhasz, Nasa Lewis Research Center; private communication (1992).